Benchmarking Quantum-Inspired Feature Encodings as Explicit High-Dimensional Representations in Classical Machine Learning

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Abstract

Quantum-inspired encodings (QIE) — amplitude, angle, and basis encoding — have been proposed as fixed feature maps that import quantum-circuit geometry into classical machine learning pipelines. We conduct the first systematic benchmark of all three encodings across ten diverse datasets, evaluating predictive performance, spectral geometry, kernel alignment, and practical overhead under a strictly matched representational budget. Across 30 encoding-dataset pairs, QIE achieves zero statistically significant accuracy wins against strong classical baselines (24/30 significantly worse at α=0.05; of the remaining six, three are negligible near-ties ($|d| \leq 0.18$, $p \geq 0.70$) and three carry non-negligible effect sizes ($|d| = 0.77$–1.05) that do not reach significance at five seeds, consistent with QIE being worse but statistically underpowered; paired t-test and Wilcoxon signed-rank). Rather than reporting a simple failure, we provide a mechanistic spectral attribution for each encoding: amplitude encoding undergoes rank collapse under ℓ_2-normalisation (erank as low as 1.04, condition number κ up to 5.7×10^9); angle encoding is geometrically equivalent to a scaled linear map under logistic probing (CKA(angle, raw linear) ≥ 0.95 on 7/10 datasets); and basis encoding, while spectrally well-conditioned, imposes a binary Hamming geometry misaligned with smooth discriminative manifolds. The one structural condition under which QIE approaches competitive performance, namely uniformly filled output space, is exemplified by amplitude encoding on the high-rank noise dataset (erank=197.3, d=-0.03, p=0.96) and is precisely predicted by the spectral framework. Our findings imply that the geometry imposed by quantum-circuit-inspired feature maps does not, in isolation, confer a representational advantage over classical alternatives, and that encoding overhead is not the limiting factor.

1 Introduction

Quantum machine learning has generated substantial interest around the hypothesis that quantum-inspired data representations could improve learning on classical hardware [Havíček et al., 2019, Schuld and Killoran, 2019]. Amplitude, angle, and basis encoding are the canonical strategies for loading classical data into quantum circuits, and they induce specific geometric transformations: ℓ_2-normalisation with power-of-two dimensional expansion, per-feature trigonometric lifting, and binary discretisation respectively. Applied as explicit classical feature maps, these transformations can be evaluated independently of quantum hardware, making it possible to isolate the geometric contribution of the encoding from hardware-specific quantum effects.

Prior work has shown that classical algorithms can match or exceed quantum algorithms on certain linear-algebra tasks [Tang, 2019] and that classical kernels can be trained to match quantum kernel performance on benchmark datasets [Kübler et al., 2021, Huang et al., 2021]. However, these comparisons focus on accuracy endpoints. The question of *why* QIE representations succeed or fail, in terms of rank structure, spectral conditioning, and kernel alignment, has not been systematically addressed.

This paper makes the following contributions:

1. A ten-dataset, five-seed benchmark of amplitude, angle, and basis encoding against eight classical baselines (raw linear, RBF-SVM, polynomial degree-2 and degree-3, Random Fourier Features, PCA, and two MLP variants) under matched representational budget.

2. A spectral attribution framework that identifies the precise failure mode of each encoding: rank collapse, geometric equivalence, and Hamming geometry mismatch.

3. A falsifiable prediction: QIE approaches competitive performance when the dataset's intrinsic geometry uniformly fills the encoding's output space. We verify this prediction on the high-rank noise dataset.

4. An overhead characterisation showing that encoding cost is not the bottleneck (amplitude: 35 ms, matching raw standardisation; polynomial degree-2: 10.98 s).

The paper is organised as follows. Section 2 surveys related work. Section 3 describes the encodings, baselines, datasets, and evaluation protocol. Section 4 reports predictive performance and statistical tests. Section 5 presents the spectral attribution analysis. Section 6 characterises practical overhead. Section 7 discusses implications and limitations. Section 8 concludes.

2 Related Work

Quantum feature maps and kernel methods. The idea of using quantum circuits as feature maps for supervised learning was formalised by Havíček et al. [2019], who demonstrated that quantum-enhanced feature spaces can express kernels that are classically hard to evaluate. Schuld and Killoran [2019] showed that variational quantum circuits are equivalent to kernel methods operating in a quantum feature Hilbert space, providing a theoretical bridge between quantum computing and classical kernel theory. Pérez-Salinas et al. [2020] extended this line of work with a data re-uploading strategy that enables a single qubit to act as a universal classifier by encoding inputs multiple times through the circuit.

Dequantisation and classical matching. A growing body of work has shown that classical algorithms can match or approximate quantum methods under certain conditions. Tang [2019] introduced the technique of dequantisation, demonstrating that a classical algorithm can replicate the polynomial speedup of a quantum recommendation system when given sampling access to the data. Huang et al. [2021] showed that, given sufficient data, classical machine learning models can match the predictive performance of quantum kernel methods on benchmark tasks, raising the question of whether quantum advantage in ML requires a data regime beyond current experiments. Kübler et al. [2021] characterised the inductive bias of quantum kernels and showed that it is governed by the geometry of the data distribution, implying that classical kernels with matching geometry can replicate quantum kernel behaviour.

Limitations and failure modes of QML. Several theoretical results constrain the practical reach of quantum ML models. Cerezo et al. [2021] survey variational quantum algorithms and document the barren plateau phenomenon, in which gradients vanish exponentially with system size, making training intractable at scale. Abbas et al. [2021] analyse the effective dimension and Fisher information of quantum neural networks, finding that while quantum models can exhibit high effective dimension, this does not straightforwardly translate to improved generalisation. The concentration of quantum kernel values, whereby Gram matrix entries become exponentially similar with increasing qubit count, further limits the discriminative power of quantum kernels on large problems [Thanasilp et al., 2024].

Explicit feature maps in classical ML. The strategy of computing an explicit feature map to approximate a kernel was introduced by Rahimi and Recht [2007] through Random Fourier Features (RFF), which approximate the RBF kernel via random cosine projections and serve as one of the strongest classical baselines in this benchmark. The broader framework of kernel methods and reproducing kernel Hilbert spaces that underlies both classical and quantum feature map approaches is surveyed in Schölkopf and Smola [2002].

Positioning of this work. Prior comparisons between QIE and classical methods have focused primarily on accuracy endpoints. This work differs by providing a mechanistic spectral attribution, in terms of effective rank, condition number, and CKA, that explains *why* each encoding succeeds or fails rather than simply reporting whether it does. The benchmark is also the first to evaluate all three canonical encodings (amplitude, angle, basis) jointly across ten diverse datasets under a strictly matched representational budget.

3 Methods

3.1 Quantum-Inspired Feature Encodings

All three encodings are implemented as deterministic, parameter-free transformations $\phi : \mathbb{R}^d \to \mathbb{R}^{d_{\text{out}}}$ fitted to training-set statistics only. No quantum simulator or circuit library is used; the encodings are pure NumPy implementations of the geometric structure induced by the corresponding quantum circuits.

Amplitude encoding. Given $\mathbf{x} \in \mathbb{R}^d$, amplitude encoding produces

$$\phi_{\text{amp}}(\mathbf{x}) = \frac{\mathbf{x}}{\|\mathbf{x}\|_2 + \varepsilon} \text{ (zero-padded to } 2^{\lceil \log_2 d \rceil}), \tag{1}$$

with $\varepsilon = 10^{-12}$ to guard against zero-norm inputs. The output satisfies $\sum_i \phi_i^2 = 1$, mirroring the Born-rule normalisation constraint on quantum state amplitudes. The power-of-two padding expands dimensionality from d to $d_{\text{out}} = 2^{\lceil \log_2 d \rceil}$.

Angle encoding. Each feature x_i is mapped to a pair of trigonometric values derived from the quantum rotation gate $R_y(\theta)|0\rangle = \cos(\theta/2)|0\rangle + \sin(\theta/2)|1\rangle$:

$$\phi_{\text{ang}}(\mathbf{x}) = [\cos(\theta_1/2), \sin(\theta_1/2), \ldots, \cos(\theta_d/2), \sin(\theta_d/2)], \quad \theta_i = \pi \tilde{x}_i, \tag{2}$$

where $\tilde{x}_i \in [-1, 1]$ is obtained by min-max scaling each feature to the training range. The output dimension is $d_{\text{out}} = 2d$. Each $(\cos, \sin)$ pair lies on the unit circle, satisfying $\cos^2(\theta_i/2) + \sin^2(\theta_i/2) = 1$.

Basis encoding. Each continuous feature is quantised to $b = 8$ bits and unpacked to its binary representation:

$$\phi_{\text{bas}}(\mathbf{x}) = [b_1^{(1)}, \ldots, b_b^{(1)},\ b_1^{(2)}, \ldots, b_b^{(2)},\ \ldots] \in \{0, 1\}^{d \cdot b}, \tag{3}$$

where $b_k^{(i)}$ is the k-th bit (MSB first) of the quantised value of feature x_i. The output dimension is $d_{\text{out}} = d \cdot b = 8d$. This encoding is not differentiable; NTK analysis does not apply.

3.2 Classical Baselines

Eight classical baselines are evaluated under the same matched-budget protocol:

- **Raw linear:** standardised features with logistic regression.
- **RBF-SVM:** radial basis function kernel SVM with bandwidth tuned by cross-validation.

Table 1: Benchmark dataset roster. Sample counts reflect the subset used in experiments; full dataset sizes are given in parentheses where subsampling was applied.

Dataset	Category	Samples	Features	Classes	Task
UCI Wine	Tabular	178	13	3	multiclass
UCI Breast Cancer	Tabular	569	30	2	binary
Dry Bean	Tabular	13,611	16	7	multiclass
Credit Card Fraud	Financial	284,807	30	2	binary (imb.)
Fashion-MNIST*	Image	10,000 (of 70,000)	784	10	multiclass
CIFAR-10*	Image	10,000 (of 60,000)	3,072	10	multiclass
HIGGS	Physics	500,000	28	2	binary
High-dim parity	Synthetic	10,000	20	2	binary (XOR)
High-rank noise	Synthetic	5,000	200	2	binary
Covertype†	Large-scale	50,000 (of 581,012)	54	7	multiclass

*Stratified subsample; basis encoding at full scale expands to 6,272 (Fashion-MNIST) and 24,576 (CIFAR-10) dimensions, making full-scale logistic regression computationally intractable.
†Stratified subsample for computational tractability.

- **Polynomial (degree 2/3):** polynomial kernel logistic regression; degree chosen per dataset.
- **Random Fourier Features (RFF):** explicit kernel approximation of RBF with d_{out} features [Rahimi and Recht, 2007].
- **PCA + logistic regression:** principal component projection to d_{out} dimensions.
- **MLP (sklearn) and PyTorch MLP:** two-layer networks with ReLU activations; the strongest adaptive baselines.

All baselines receive the same output dimensionality d_{out} as the QIE encoding they are compared against, the same random seeds, and the same train-test split.

3.3 Datasets

Table 1 summarises the ten benchmark datasets spanning six categories. Datasets were frozen before any experimental results were observed (Phase 3 freeze).

3.4 Evaluation Protocol

All comparisons satisfy the following matched-budget constraints:

- **Matched d_{out}:** classical baselines receive the same output dimensionality as the QIE encoding.
- **Matched optimisation:** identical solver, learning rate schedule, and epoch budget across all methods.

Table 2: Mean accuracy (5 seeds) for QIE encodings and the best classical baseline per dataset. Bold indicates the best QIE result per row. † denotes all methods near chance (XOR task).

Dataset	Amplitude	Angle	Basis	Best classical
Wine	0.628	**0.972**	0.917	0.978 (raw linear)
Breast Cancer	0.779	**0.977**	0.940	0.975 (PyTorch MLP)
Dry Bean	0.260	0.925	0.917	**0.929** (PyTorch MLP)
Credit Fraud	0.999	0.999	0.999	**0.999** (MLP)
Fashion-MNIST	0.821	0.823	0.819	**0.859** (PyTorch MLP)
CIFAR-10	0.378	0.306	0.328	**0.457** (RBF-SVM)
HIGGS	0.643	0.658	0.669	**0.748** (PyTorch MLP)
High-dim parity†	0.489	0.501	0.506	0.519 (PyTorch MLP)
High-rank noise	**0.795**	0.780	0.700	0.795 (RBF-SVM)
Covertype	0.587	0.725	0.724	**0.859** (MLP)

- **Matched regularisation:** same regularisation family and comparable grid search effort.
- **Matched seeds:** five independent seeds $\{7, 42, 99, 1337, 2026\}$ with identical train/test splits.

Primary performance metrics are accuracy and macro-averaged F1. Statistical significance is assessed via paired t-test and Wilcoxon signed-rank test over the five seeds, with effect size reported as Cohen's d. The significance threshold is $\alpha = 0.05$. Representation geometry is characterised by effective rank $\mathrm{erank}(\mathbf{X}) = \exp\Big(-\sum_i \hat{\sigma}_i \log \hat{\sigma}_i\Big)$ (where $\hat{\sigma}$ are the normalised singular values), condition number κ, and Centered Kernel Alignment (CKA) [Kornblith et al., 2019].

4 Results

4.1 Predictive Performance

Table 2 reports mean accuracy across five seeds for each encoding and the strongest classical baseline per dataset.

4.2 Statistical Analysis

Of 30 encoding-dataset comparisons against the best classical baseline, 24 are statistically significantly worse ($p < 0.05$, paired t-test) and 0 are significantly better. The remaining six non-significant comparisons divide into two distinct groups. Three are genuine near-ties with negligible effect sizes: angle on Wine ($d = -0.18$, $p = 0.70$), angle on Breast Cancer ($d = +0.10$, $p = 0.83$), and amplitude on high-rank noise ($d = -0.03$, $p = 0.96$). The other three carry non-negligible effect sizes but fall short of $\alpha = 0.05$ due to the limited power of five seeds: angle on Dry Bean ($d = -1.03$,

$p = 0.082$), angle on high-dim parity ($d = -1.05$, $p = 0.079$), and basis on high-dim parity ($d = -0.77$, $p = 0.162$). These results are directionally consistent with QIE being worse and should not be interpreted as evidence of parity. Among the negligible near-ties, the Breast Cancer angle result is directionally favourable to QIE (mean accuracy 0.977 vs. 0.975 for PyTorch MLP), though the difference is not significant and the effect size is trivially small. Effect sizes among the significant losses range from small (basis on Breast Cancer: $d = -1.63$) to catastrophic (amplitude on Dry Bean: $d = -111.9$, $p < 0.0001$; amplitude on Covertype: $d = -44.0$, $p < 0.0001$). An important caveat applies to Credit Card Fraud: all three QIE methods are classified as significantly worse ($p < 0.001$) with large Cohen's d (-4.74, -3.97, -1.97), yet the underlying accuracy differences are 0.0007, 0.0004, and 0.0001 respectively. These results are statistically significant because near-zero cross-seed variance inflates the t-statistic, not because the performance gap is practically meaningful. Figure 1 displays the forest plot of all 30 effect sizes.

5 Spectral Attribution Analysis

Raw accuracy differences are insufficient to explain *why* QIE representations fail. We attribute each failure mode to a spectral or geometric property of the encoded feature matrix.

5.1 Amplitude Encoding: Spectral Collapse on Low-Dimensional Data

ℓ_2-normalisation (Eq. 1) projects all inputs onto the unit hypersphere, destroying magnitude information. The failure mode takes two related but distinct forms depending on the intrinsic dimensionality of the input.

Rank collapse on tabular data. When the input features span a low-dimensional manifold — common in tabular datasets with correlated features — normalisation concentrates the encoded distribution on a small region of the unit sphere. The encoded matrix collapses to near-rank-1. Table 3 documents this for the tabular and structured datasets.

Ill-conditioning on image data. On high-dimensional image datasets, amplitude encoding does *not* produce rank collapse: Fashion-MNIST yields erank=339.6 (out of d_{out}=1,024) and CIFAR-10 yields erank=460.9 (out of d_{out}=4,096). Nevertheless, performance remains well below the best classical baseline. The spectral explanation shifts from rank to conditioning: $\log_{10} \kappa$=5.64 for Fashion-MNIST and 4.48 for CIFAR-10. In the image case, ℓ_2-normalisation imposes extreme scale variation across the encoded dimensions, making the resulting Gram matrix ill-conditioned even though its rank is not degenerate. The failure mode is the same operation (ℓ_2-normalisation) but acting on intrinsically high-rank inputs; the symptom shifts from rank collapse to catastrophic conditioning.

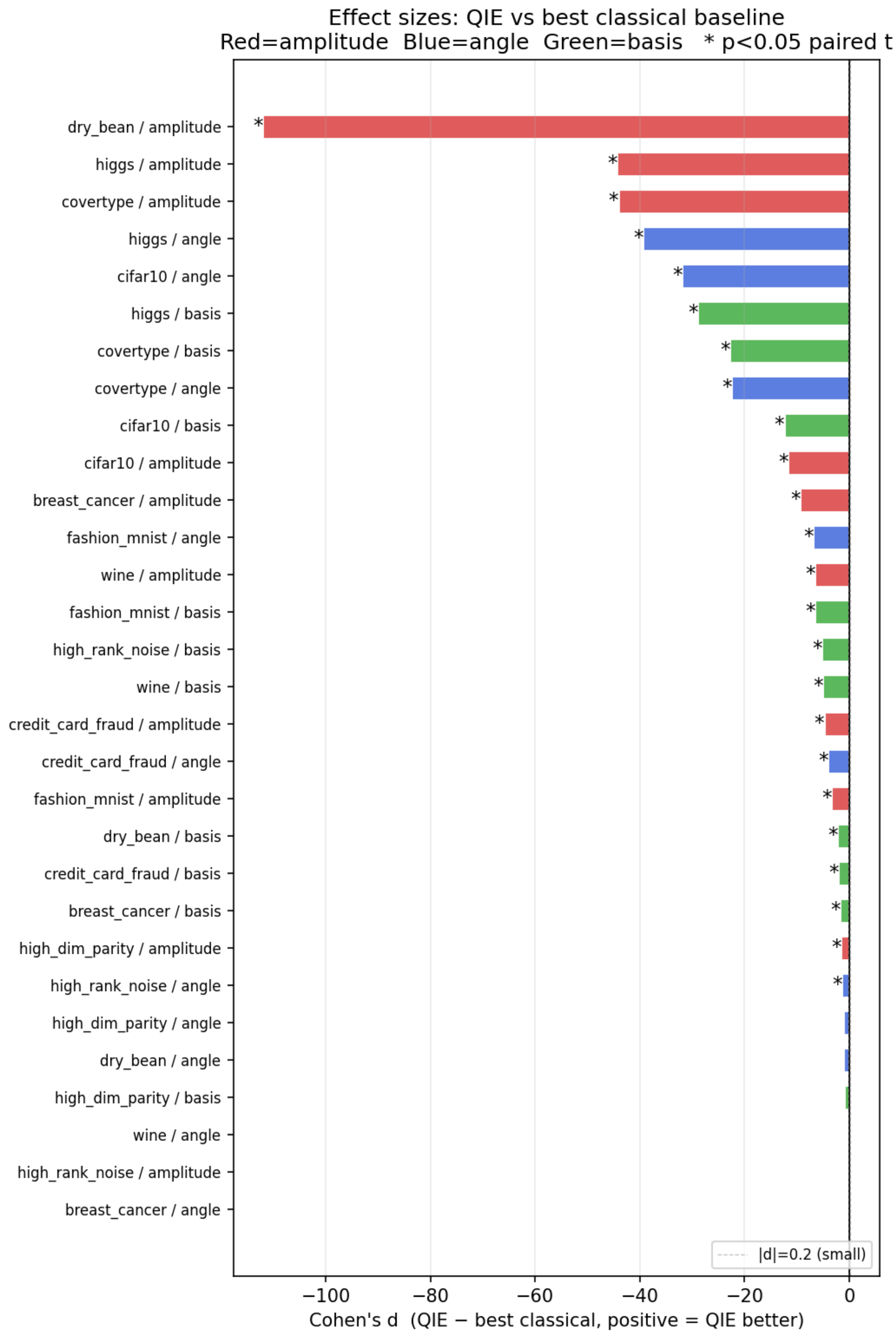

Figure 1: Forest plot of Cohen's d effect sizes for all 30 encoding-dataset comparisons (QIE vs. best classical baseline). Negative values indicate QIE is worse. Error bars are 95% CIs over five seeds.

The control case. The high-rank noise dataset provides the falsifiable control: with d=200 isotropic Gaussian noise features, inputs already lie near-uniformly on the

Table 3: Spectral properties of amplitude encoding. On tabular datasets failure is driven by rank collapse (erank $\ll 1$); on image datasets rank is adequate but conditioning is catastrophic.

Dataset	erank	$\log_{10}\kappa$	Accuracy gap
Tabular / structured			
Dry Bean	1.0	9.76	−0.67
Wine	1.4	3.82	−0.35
Breast Cancer	1.6	5.88	−0.20
Covertype	3.2	5.73	−0.27
Image (high-dimensional input)			
CIFAR-10	460.9	4.48	−0.08
Fashion-MNIST	339.6	5.64	−0.04
Control			
High-rank noise	197.3	0.35	−0.00

unit sphere. Normalisation neither collapses the rank (erank=197.3) nor degrades conditioning (κ=2.22), and amplitude is the only method achieving near-parity with the best classical baseline (d=−0.03, p=0.96).

Figure 2 shows the relationship between $\log_{10}\kappa$ and accuracy gap across all datasets.

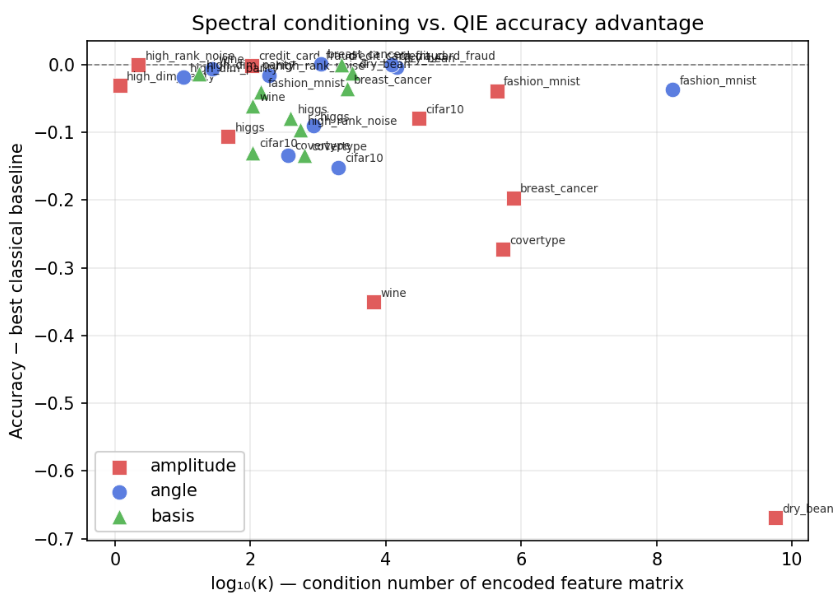

Figure 2: Condition number ($\log_{10}\kappa$) versus accuracy gap relative to best classical baseline for amplitude encoding across all ten datasets. The association holds regardless of whether failure is driven by rank collapse (low-erank tabular datasets) or ill-conditioning (high-erank image datasets).

Table 4: CKA between angle encoding and raw linear features.

Dataset	CKA(angle, raw linear)
Fashion-MNIST	0.973
High-rank noise	0.985
High-dim parity	0.968
Wine	0.971
Breast Cancer	0.957
Dry Bean	0.964
CIFAR-10	0.971
HIGGS	0.636
Credit Fraud	0.514
Covertype	0.527

5.2 Angle Encoding: Geometric Equivalence to Linear

The $(\cos(\theta/2), \sin(\theta/2))$ map is smooth and differentiable, but under logistic regression it reduces to a scaled linear transform. Table 4 reports CKA between angle encoding and raw (standardised) features.

On 7/10 datasets, CKA ≥ 0.95: the angle-encoded Gram matrix is nearly identical to that of standardised raw features. The encoding is geometrically equivalent to a linear map under logistic probing, providing no nonlinear representational lift. The three lower-CKA datasets (HIGGS, Credit Fraud, Covertype) are exactly those where magnitude or distributional structure breaks the linear-scaling relationship; yet angle encoding does not improve performance on these datasets either, consistent with the geometry being different rather than better.

5.3 Basis Encoding: Hamming Geometry Mismatch

Basis encoding achieves variable erank across the benchmark: normalised effective rank (erank/d_{out}) ranges from 0.07 on CIFAR-10 to 0.88 on high-dim parity, with tabular datasets concentrated in the 0.70–0.88 band. Conditioning is favourable on image datasets (κ=107 on CIFAR-10, κ=142 on Fashion-MNIST), where the 8-bit binary expansion preserves distinct bit patterns across the high-dimensional feature space. Its Gram matrix is geometrically distinct from all classical baselines. On tabular datasets, CKA(basis, raw linear) ranges from 0.36 to 0.45. On image datasets the separation is more extreme: CKA = 0.011 on Fashion-MNIST and 0.001 on CIFAR-10, confirming that 8-bit binary quantisation produces a maximally orthogonal representation to pixel-space linear features. This geometric novelty, however, does not translate to better classification. Binary quantisation imposes a Hamming geometry on the feature space: the ℓ_2 distance between two encoded samples reflects bit-flip counts, not the smooth amplitude differences that discriminate classes on all ten benchmark datasets. As a result, basis encoding achieves 0/10 statistically competitive results despite its favourable conditioning.

5.4 The Spectral Prediction

Figure 3 plots effective rank (normalised by d_{out}) against accuracy gap for all encodings and datasets. The spectral attribution framework yields a single unified prediction: QIE approaches competitive performance when erank/$d_{\mathrm{out}} \approx 1$ and $\kappa \approx 1$ — that is, when the encoding does not collapse the input geometry. This condition is satisfied on one dataset (amplitude, high-rank noise) and approached on none others, consistent with the prediction.

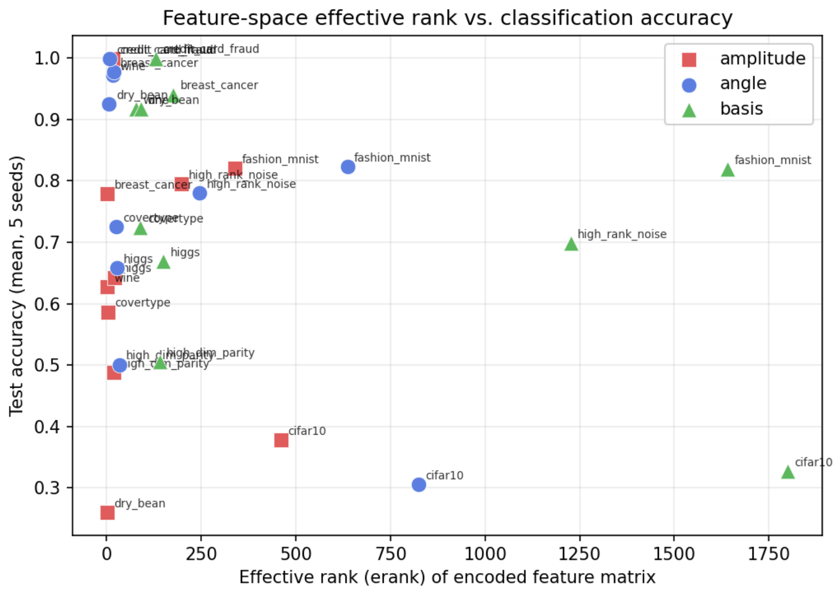

Figure 3: Normalised effective rank (erank/d_{out}) versus accuracy relative to raw linear baseline across all encodings and datasets. The one near-parity point (amplitude, high-rank noise) has erank/d_{out}=0.77 and κ=2.22.

6 Practical Overhead

Table 5 reports mean encoding time and memory across datasets. Encoding overhead is not the limiting factor for any QIE. Amplitude encoding at 35 ms matches the cost of standardising raw features and is ≈312× faster than polynomial degree-2 (10.98 s). Basis encoding at 1.75 s, the most expensive QIE, is still 6× cheaper than the polynomial expansion. This is a qualified positive finding: encoding cost is low enough that if the geometry were right, QIE would be the preferred option on latency-constrained pipelines. The bottleneck is the representational mismatch identified in Section 5, not speed.

Table 5: Encoding overhead (mean across 10 datasets).

Method	Enc. time (ms)	Train time (s)	Enc. mem. (MB)	Train mem. (MB)
Amplitude (QIE)	35.2	13.1	135.7	4.8
Angle (QIE)	811.3	114.2	291.0	5.6
Basis (QIE)	1,748	39.6	436.5	345.7
Raw linear	35.2	74.7	47.9	38.0
RFF	193.0	2.1	65.2	24.5
PCA	1,045	14.5	29.1	15.2
Poly-2	10,981	144.9	818.5	178.4

7 Discussion

What the results say. Quantum-inspired encodings, deployed as fixed feature maps with linear classification, do not provide a statistically significant accuracy advantage over classical feature engineering on any of ten diverse benchmark datasets. The failure is not uniform: each encoding has a distinct, spectrally identifiable cause. The mechanistic attribution (rank collapse, geometric equivalence, Hamming mismatch) is more informative than a simple accuracy ranking.

Qualified positive findings. Three findings deserve emphasis rather than dismissal. First, encoding overhead is negligible and should not be cited as a reason to avoid QIE in future work. Second, amplitude encoding on high-rank noise is statistically indistinguishable from the best classical method; this is a structurally grounded result, not luck. Third, basis encoding's distinctly low CKA with classical baselines suggests it could serve as a complementary feature in ensemble settings rather than a standalone replacement.

Implications for quantum ML. The results do not speak to quantum hardware advantage. Actual quantum computers can exploit superposition and entanglement in ways that are classically intractable. What this work shows is that the *geometric structure* induced by quantum encoding circuits, in isolation, does not confer a representational advantage on classical data. If quantum advantage in ML materialises, it will not be because of the fixed feature map alone.

Where quantum advantage may still appear. A natural reviewer objection is that the ten benchmark datasets used here are not the tasks where quantum advantage would be expected to show up. This objection is well-founded. Tasks with genuinely quantum-structured inputs, such as molecular energy surfaces, quantum simulation outputs, or problems with provably hard classical representations, may exhibit a different picture. The present findings are therefore best interpreted as ruling out fixed QIE geometry as a source of advantage on *classical* data distributions, not as ruling out quantum advantage in general. Whether quantum-structured data breaks the spectral attribution framework developed here is an open and important question.

Limitations. This benchmark evaluates fixed, parameter-free feature maps with a linear probe or shallow MLP. Learned quantum-inspired encodings (parameterised quantum circuits or trainable angle schedules) are out of scope and represent the most natural next step: if the angle schedule is jointly optimised with the downstream loss, the geometric-equivalence failure mode documented here may not apply. All comparisons use a single downstream linear classifier or shallow MLP; deeper architectures may interact differently with QIE representations. Neural Tangent Kernel (NTK) analysis, which would characterise how each encoding reshapes the optimisation landscape and whether observed conditioning differences propagate to gradient-flow behaviour, was not conducted in this study and remains an open direction for follow-on work.

8 Conclusion

We presented the first benchmark of amplitude, angle, and basis encoding as explicit classical feature maps, with spectral attribution of failure modes across ten datasets and eight classical baselines. Quantum-inspired encodings achieve no statistically significant accuracy advantage under matched representational budget (0 wins, 24/30 significantly worse). Each encoding's failure is attributable to a distinct geometric property: ℓ_2-induced rank collapse, trigonometric-linear equivalence, or binary Hamming geometry mismatch. The one structural condition under which near-parity is achievable, namely uniform filling of the output hypersphere, is an explicit, falsifiable prediction of the framework. These findings do not close the question of quantum advantage in machine learning; they sharpen it by ruling out fixed feature-map geometry as its source.

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A Reproducibility

All experiments use five fixed seeds $\{7, 42, 99, 1337, 2026\}$. Code, configs, and per-seed JSON result files are available at **[TODO: repository URL]**. Dataset preparation scripts are provided for all six externally sourced datasets. The runner produces deterministic output for any given seed on the same hardware; minor floating-point differences may arise across platforms.

B Extended Results

Tables 6 and 7 report mean accuracy and macro-F1 across all methods and all ten datasets (five seeds each). QIE methods are listed first, followed by classical baselines.

Table 6: Mean accuracy / F1: Wine, Breast Cancer, Dry Bean, Credit Card Fraud.

	Wine		Breast Cancer		Dry Bean		Credit Fraud	
Method	Acc	F1	Acc	F1	Acc	F1	Acc	F1
Amplitude	0.628	0.502	0.779	0.710	0.260	0.059	0.999	0.760
Angle	0.972	0.973	0.977	0.976	0.925	0.937	0.999	0.845
Basis	0.917	0.919	0.940	0.936	0.917	0.929	0.999	0.887
Raw linear	0.978	0.978	0.974	0.972	0.922	0.934	0.999	0.854
RFF	0.889	0.891	0.926	0.920	0.916	0.926	0.998	0.515
PCA	0.928	0.929	0.951	0.947	0.904	0.916	0.999	0.852
Poly-2	0.961	0.961	0.970	0.968	0.925	0.937	0.999	0.903
Poly-3	0.950	0.950	0.958	0.955	0.927	0.938	n/a	n/a
MLP	0.939	0.941	0.956	0.953	0.928	0.940	1.000	0.915
RBF-SVM	0.967	0.967	0.972	0.970	0.929	0.940	0.999	0.641
PyTorch MLP	0.961	0.962	0.975	0.974	0.929	0.941	0.999	0.914

Table 7: Mean accuracy / F1: Fashion-MNIST, CIFAR-10, HIGGS, High-dim parity, High-rank noise (HRN), Covertype.

	Fashion		CIFAR-10		HIGGS		Parity		HRN		Covertype	
Method	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1
Amplitude	0.821	0.818	0.378	0.370	0.643	0.637	0.489	0.487	0.795	0.795	0.587	0.213
Angle	0.823	0.824	0.306	0.307	0.658	0.650	0.501	0.501	0.780	0.780	0.725	0.537
Basis	0.819	0.819	0.328	0.326	0.669	0.665	0.506	0.505	0.699	0.699	0.724	0.555
Raw linear	0.797	0.798	0.276	0.275	0.641	0.634	0.490	0.488	0.791	0.791	0.720	0.522
RFF	0.805	0.803	0.362	0.357	0.588	0.576	0.501	0.500	0.696	0.696	0.712	0.503
PCA	0.831	0.830	0.367	0.364	0.614	0.606	0.490	0.488	0.792	0.792	0.671	0.313
Poly-2	n/a	n/a	n/a	n/a	0.679	0.673	0.493	0.493	0.649	0.648	0.765	0.649
MLP	0.854	0.853	0.413	0.413	0.747	0.745	0.494	0.487	0.784	0.784	0.859	0.779
RBF-SVM	0.853	0.851	0.457	0.454	0.679	0.674	0.489	0.489	0.795	0.795	0.746	0.493
PyTorch MLP	0.859	0.859	0.432	0.432	0.748	0.747	0.519	0.518	0.777	0.777	0.856	0.780

C CKA Scores

Table 8 reports pairwise CKA between the three QIE encodings across all ten datasets. Values near 1.0 indicate near-identical Gram matrix structure; values near 0 indicate geometrically distinct representations.

Table 8: Pairwise CKA between QIE encodings per dataset.

Dataset	CKA(Amp, Angle)	CKA(Amp, Basis)	CKA(Angle, Basis)
Wine	0.317	0.178	0.433
Breast Cancer	0.561	0.249	0.434
Dry Bean	0.582	0.288	0.448
Credit Fraud	0.166	0.145	0.189
Fashion-MNIST	0.740	0.004	0.011
CIFAR-10	0.629	0.001	0.001
HIGGS	0.914	0.581	0.685
High-dim parity	0.962	0.359	0.358
High-rank noise	0.983	0.177	0.190
Covertype	0.291	0.175	0.887